

Fuzzy membership functions

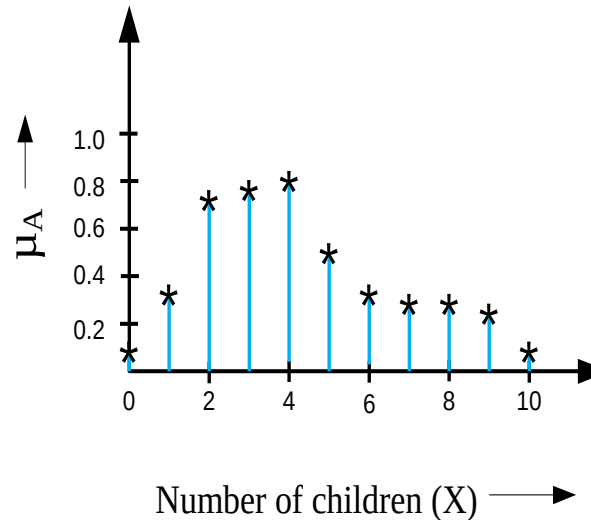
Fuzzy membership functions

A fuzzy set is completely characterized by its membership function (sometimes abbreviated as MF and denoted as μ). So, it would be important to learn how a membership function can be expressed (mathematically or otherwise).

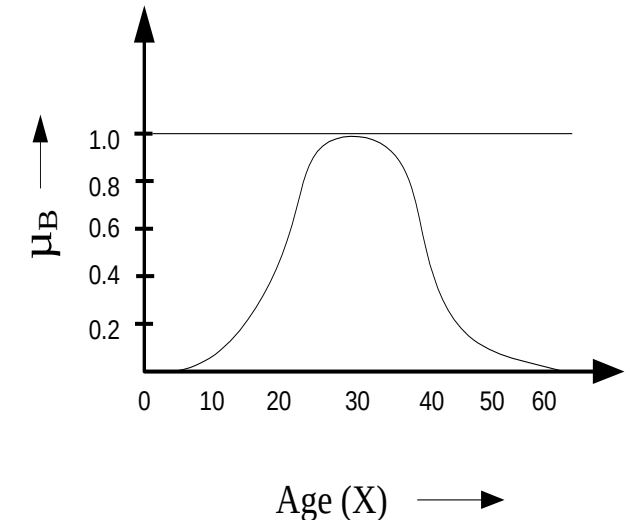
Note: A membership function can be on

a) a discrete universe of discourse and

b) a continuous universe of discourse.



A = Fuzzy set of "Happy family"

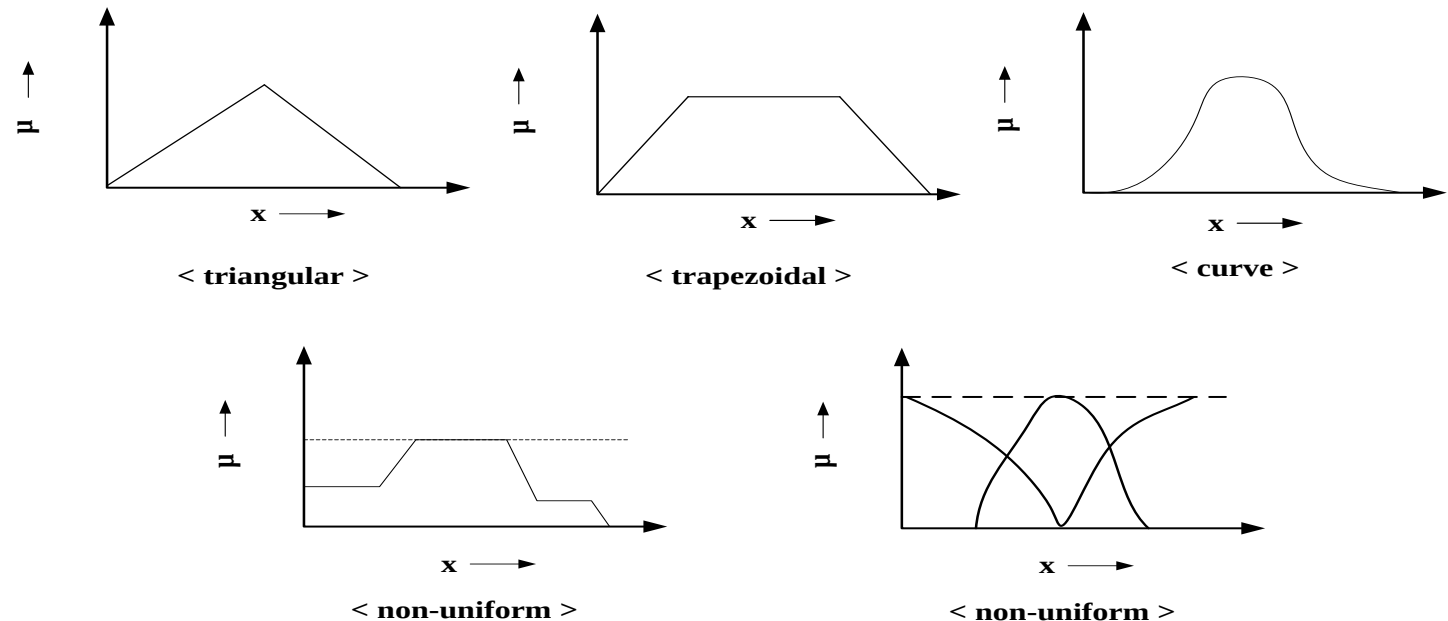


B = "Young age"

Fuzzy membership functions

So, membership function on a discrete universe of course is trivial. However, a membership function on a continuous universe of discourse needs a special attention.

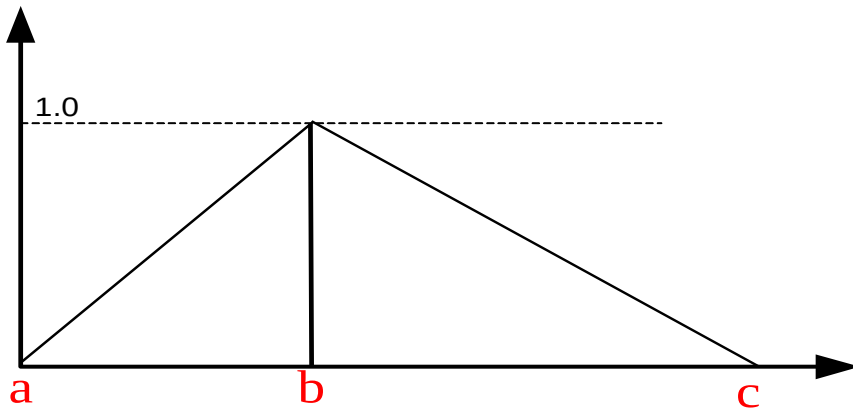
Following figures shows typical examples of membership functions.



Fuzzy MFs : Formulation and parameterization

In the following, we try to parameterize the different MFs on a continuous universe of discourse.

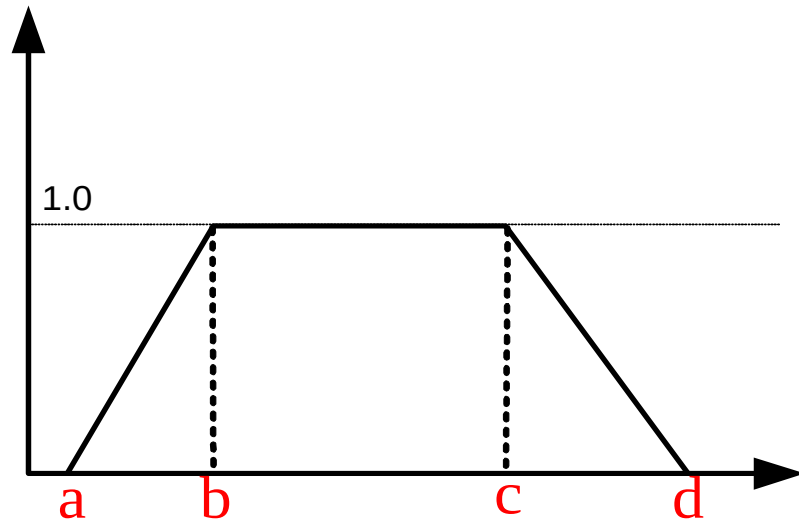
Triangular MFs : A triangular MF is specified by three parameters $\{a, b, c\}$ and can be formulated as follows.



$$\text{triangle}(x; a, b, c) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x - a}{b - a} & \text{if } a \leq x \leq b \\ \frac{c - x}{c - b} & \text{if } b \leq x \leq c \\ 0 & \text{if } c \leq x \end{cases}$$

Fuzzy MFs: Trapezoidal

A trapezoidal MF is specified by four parameters $\{a, b, c, d\}$ and can be defined as follows:

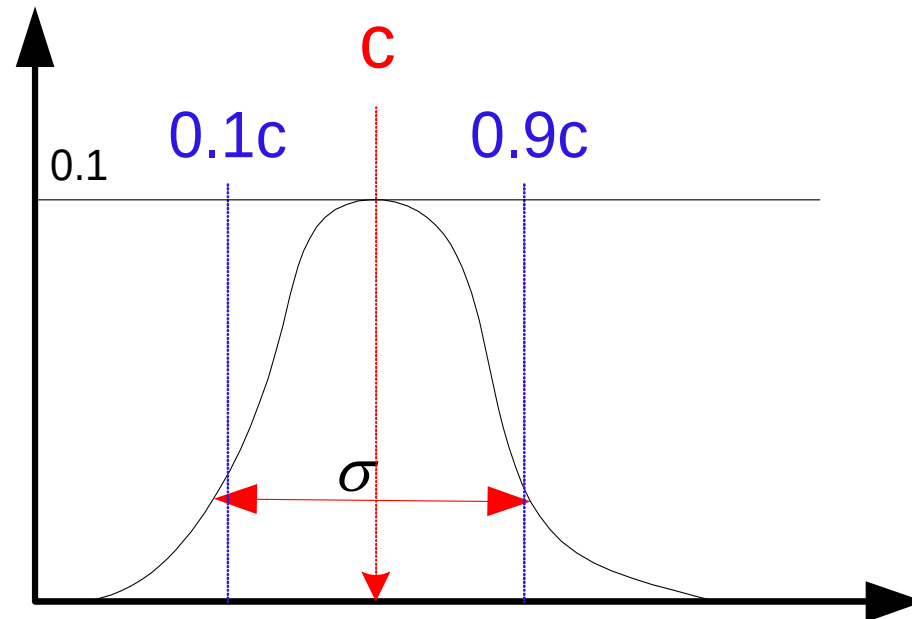


$$\text{trapezoid}(x; a, b, c, d) = \begin{cases} 0 & \text{if } x \leq a \\ \frac{x - a}{b - a} & \text{if } a \leq x \leq b \\ 1 & \text{if } b \leq x \leq c \\ \frac{d - x}{d - c} & \text{if } c \leq x \leq d \\ 0 & \text{if } d \leq x \end{cases}$$

Fuzzy MFs: Gaussian

A **Gaussian MF** is specified by two parameters $\{c, \sigma\}$ and can be defined as below:

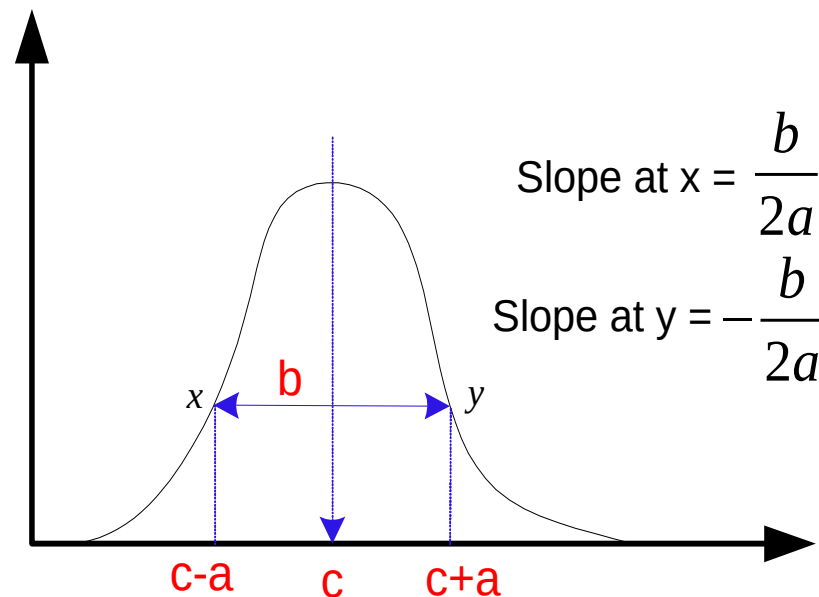
$$\text{gaussian}(x; c, \sigma) = e^{-\frac{1}{2}\left(\frac{x-c}{\sigma}\right)^2}$$



Fuzzy MFs: Generalized bell

It is also called **Cauchy MF**. A generalized bell MF is specified by three parameters $\{a, b, c\}$ and is defined as:

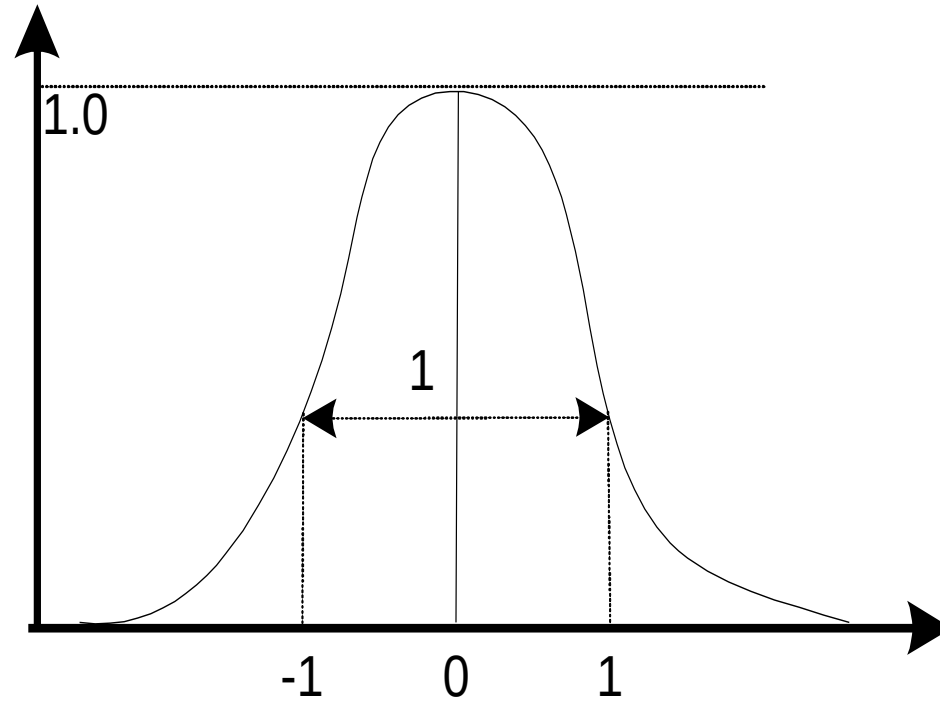
$$bell(x; a, b, c) = \frac{1}{1 + \left| \frac{x - c}{a} \right|^{2b}}$$



Example: Generalized bell MFs

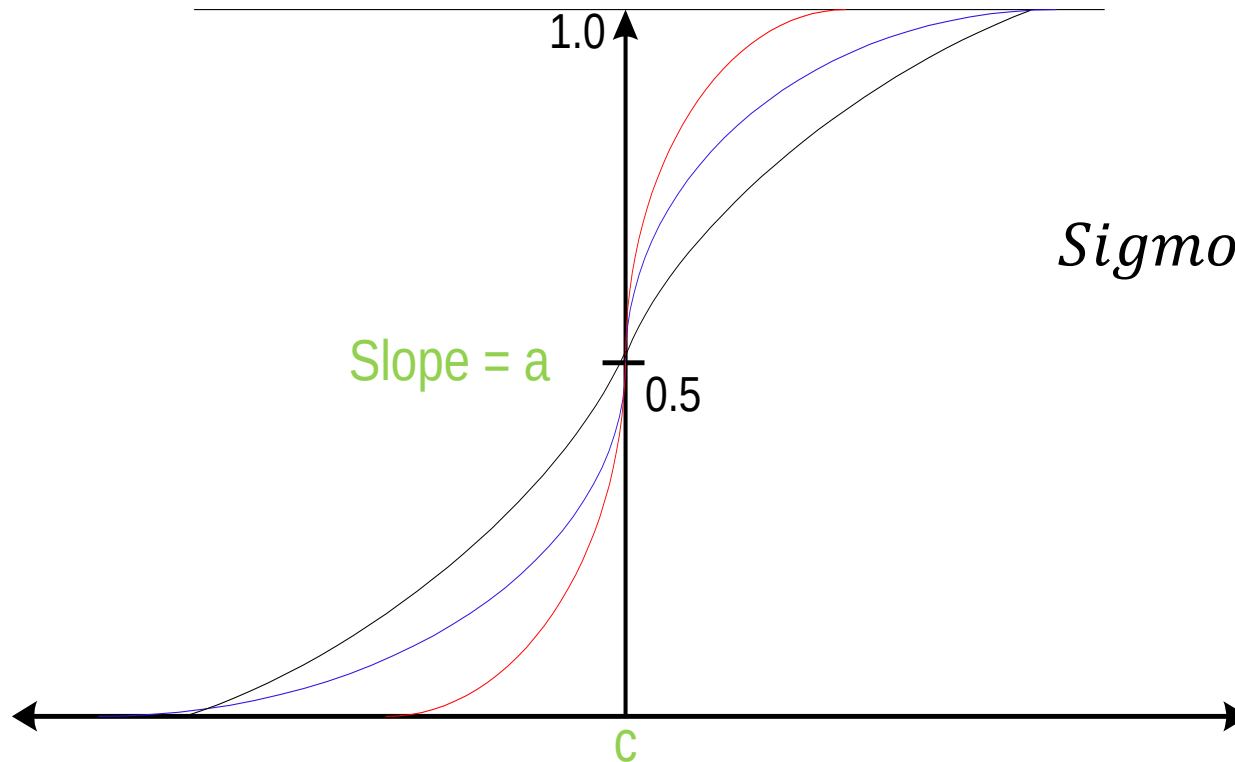
Example: $\mu(x) = \frac{1}{1+|x|^2};$

$a = b = 1$ and $c = 0;$



Fuzzy MFs: Sigmoidal MFs

Parameters: $\{a, c\}$; where c = crossover point and a = slope at c ;



$$\text{Sigmoid}(x; a, c) = \frac{1}{1 + e^{-\left[\frac{x-c}{a}\right]}}$$